

Education

Aug 2025-Present	MBZUAI · Ph.D. in Machine Learning	Advisor: Dr. Zhiqiang Shen
Sep 2021-Apr 2025	Technical University of Denmark · B.Sc. in General Engineering	Advisor: Prof. Dimitrios Papadopoulos
Sep 2020-Apr 2021	University of Edinburgh · B.Sc. in Mathematics	

Research Interest

I work on post-training agents for long, expert work: how to get agents to do it, and how to keep them getting better at it. My answer is to post-train the model and co-evolve the harness around it. My long-term goal is agentic systems that can carry out arbitrarily complex, long-horizon human expert work in digital environments, with a path toward real-world tasks. My current focus is artifact design: production-grade models for agentic design, and automated iteration systems for the long-horizon harnesses that support them.

Research Experience

Meituan · M17 LongCat Research Intern, Northern Dipper Talent Program Sep 2026-Present

- Research and build Agent Harnesses for **long-horizon multimodal design tasks**, producing trajectories for subsequent model training and integrating execution, tool use, trace capture, automated evaluation, and iterative optimization into a reproducible pipeline.
- Investigate **harness automation and meta-optimization**: within a fixed model capability boundary, use rollout evidence, evaluator feedback, and component-level updates to improve reliable completion of long-horizon tasks.
- Contribute to the full **LongCat 2.5 post-training workflow for Agentic Design**, from post-training harness design and training-oriented trajectory collection to **Mid-Training and subsequent SFT and RFT**, studying how to internalize harness behaviors and trajectories into model parameters; explored **direct rewriting, hint-based control, teacher continuation from student steps, and RL**, with hands-on insights into each.

Meituan · M17 LongCat Research Intern Apr 2026-Sep 2026

- Focused on **AutoDesign** to explore long-horizon design agents and harness methods, validating an end-to-end research loop spanning reusable DesignHarnesses, evaluation protocols, and real deliverables, and establishing a systems foundation for subsequent foundation-model agent and harness iteration.

Mohamed bin Zayed University of Artificial Intelligence (MBZUAI) Research Assistant Jan-Aug 2025

- Studied how language-pretrained parameters transfer and generalize to vision-only tasks; developed label-free random-label bridge training for cross-modal parameter adaptation and foundation-capability analysis.
- Built Open CaptchaWorld, an interactive benchmark for multimodal browser agents spanning 20 categories and 225 dynamic CAPTCHAs, quantifying the long-horizon interaction gap between humans and frontier agents.

Open-Source Agent and Harness Systems

AutoDesign 238 Stars · 2026.10.09 2026-Present

- Built an open-source Design Agent and model-agnostic harness for academic communication artifacts, transforming a paper into an editable poster, slide deck, webpage, and subtitled video; supports coding harnesses including Codex, Claude Code, and Kimi Code.
- Proposed **Meta-Harness Optimization**: an inner Designer-Critic loop iterates native artifacts, while an outer loop updates reusable DesignHarnesses from traces, evaluators, and persistent optimization records, with acceptance gates preventing regressions.
- Built PosterBench over 100 papers. In public results, AutoDesign reached **78.32**, outperforming Claude Design by **7.45** points and delivering **+5.01 to +19.56** point gains across seven matched configurations.

github.com/Yaxin9Luo/AutoDesign

FigMirror 523 Stars · 2026.10.09 2026-Present

- Contributed to a reference-image-driven scientific figure agent that combines Orchestrator-Drawer-Reviewer roles with Grounded Measurement to turn user data into editable Matplotlib code and camera-ready PDFs.
- Authored early specifications for the method, Web UI, and multi-round refinement; added Codex and Claude runners, event-bus and session-state coverage, and refinement integration tests to establish a verifiable baseline for subsequent agent iterations.

github.com/VILA-Lab/FigMirror

First- and Co-First-Author Publications

01 AutoDesign: Meta-Harness Optimization for Long-Horizon Agentic Design

First author · arXiv 2026 · Long-Horizon Agentic Design

Introduced Meta-Harness Optimization, which recursively improves reusable DesignHarnesses using rollouts, evaluation, and acceptance gates. AutoDesign scored 78.32 on PosterBench, 7.45 points above Claude Design under the same configuration, and was featured by Synced.

arxiv.org/abs/2608.13560 · Synced feature

02 Next-Gen CAPTCHAs: Leveraging the Cognitive Gap for Scalable and Diverse GUI-Agent Defense

Co-first author · ICML 2026 · GUI-Agent Defense

Developed a scalable GUI-agent defense framework around the human-agent cognitive gap, designing dynamic tasks that probe differences in interactive perception, memory, decision-making, and action. A data-generation pipeline and backend-supported task types continuously produce diverse CAPTCHA instances at scale.

arxiv.org/abs/2602.09012

03 Open CaptchaWorld: A Comprehensive Web-based Platform for Testing and Benchmarking Multimodal LLM Agents

First author · NeurIPS 2025 · Multimodal Agents

Introduced the first dynamic web CAPTCHA benchmark for multimodal LLM agents, covering 20 categories, 225 tasks, and the CAPTCHA Reasoning Depth metric. Humans achieved a 93.3% success rate, compared with 40.0% for the strongest evaluated agent.

arxiv.org/abs/2505.24878

04 γ -MoD: Exploring Mixture-of-Depth Adaptation for Multimodal Large Language Models

First author · ICLR 2025 · Efficient Multimodal Foundation Models

Introduced a Mixture-of-Depth adaptation strategy for existing MLLMs that reduces serving cost through selective token and layer activation while avoiding the substantial degradation caused by directly converting dense layers to MoD.

openreview.net/forum?id=1EnpStvBU8

05 APL: Anchor-based Prompt Learning for One-stage Weakly Supervised Referring Expression Comprehension

First author · ECCV 2024 · Vision-Language Alignment

Designed an Anchor-based Prompt Encoder driven by position, color, and category prompts, together with text-reconstruction and visual-alignment objectives. The method achieved state-of-the-art results on four REC benchmarks, improving over RefCLIP by 6.44% on RefCOCO.

ecva.net · ECCV 2024 / APL

Collaborative Publications

1. [arXiv 2026](#) · VideoCoCo: Code-as-CoT for Physically-Consistent Video Generation via an Agentic Dual-Engine System
2. [ACL 2025](#) · DRAG: Distilling RAG for SLMs from LLMs via Evidence and Graph-based Distillation
3. [CVPR 2025](#) · DViN: Dynamic Visual Routing Network for Weakly Supervised Referring Expression Comprehension

Additional Experience

- IEEE Cybermatics Congress 2024: volunteer organizing-team member; served as conference assistant and session chair for the Smart Data workshop.
- SciSec 2024: conference volunteer. Institute of Social Science Survey, Peking University: Summer Intern, Public Mental Health Project (Jul-Sep 2019).